Contents

1 Introduction

Epilepsy is a chronic neurological disorder characterized by recurrent, transient episodes of abnormal, synchronous neuronal activity known as seizures. Globally, epilepsy affects approximately 65 million individuals and imposes a substantial social and economic burden, including increased disability and mortality rates (Leonardi and Ustun 2002; Feigin et al. 2021). For the roughly one-third of patients with drug-resistant epilepsy, the unpredictability of seizure occurrence presents a significant challenge in daily life (Schulze-Bonhage and Kühn 2008). The development of reliable, automated seizure prediction algorithms holds the potential to improve patient quality of life by enabling personalized interventions, such as adaptive medication regimens, closed-loop neurostimulation, or timely precautionary actions by patients themselves, thereby increasing patient empowerment and reducing uncertainty.

1.1 Epileptic Cycles and the Proictal State

In seizure prediction, we distinguish between "preictal" states—periods preceding a seizure—and "interictal" states, which are temporally distant from any seizure event (Yang et al. 2024). Numerous studies have demonstrated that machine learning models can distinguish these states with performance significantly better than chance in both human and canine intracranial EEG (iEEG) recordings (Brinkmann et al. 2016; Kuhlmann, Karoly, et al. 2018). Notably, Cook et al. (2013) provided evidence that, in some cases, seizures can be predicted hours in advance.

A growing body of research has revealed that seizures and related interictal epileptiform activity (IEA) are not randomly distributed in time, but instead exhibit rhythmicity across multiple temporal scales, including circadian (daily), multidien (multi-day), and circannual (yearly) cycles (Karoly et al. 2021). Seizure occurrence is often phase-locked to specific phases of these cycles, suggesting that the brain's susceptibility to seizures fluctuates over time.

Despite these insights, the mechanisms underlying ictogenesis—the process by which a normal brain transitions into a seizure—remain incompletely understood (Kuhlmann, Lehnertz, et al. 2018). A central question is whether seizures arise spontaneously, with little or no warning (i.e., within seconds to milliseconds), or whether the brain undergoes a gradual transition into a state of heightened seizure susceptibility over minutes, hours, or even longer. The latter scenario is conceptualized as the "proictal" state, in which the brain is primed for seizure generation and only a minor perturbation is required to trigger a seizure. In this framework, certain phases of the aforementioned cycles correspond to periods of increased proictal risk.

In the context of seizure prediction, two major challenges persist: (1) understanding if or why some seizures are not preceded by detectable preictal changes and thus remain unpredictable (sensitivity), and (2) elucidating the origins of false positive predictions (specificity). One hypothesis is that not all seizures are preceded by reliable preictal signatures, reflecting truly spontaneous ictogenesis. Alternatively, limitations in model architecture or feature selection may prevent the detection of existing preictal patterns. A further possibility is that false positive predictions may, in fact, correspond to periods when the brain is in a proictal state—i.e., a seizure was likely, but the critical perturbation did not occur, resulting in a subclinical event (Müller et al. 2022).

1.2 Research Approach

In this work, we systematically investigate these possibilities. Two distinct model types are trained: a convolutional neural network (CNN) that processes raw iEEG signals, and an ensemble of multilayer perceptrons that operate on hand-crafted features extracted from the signals. The temporal co-occurrence of false positive predictions across these models is interpreted as evidence for the existence of a proictal state. Additionally, we analyze whether false positive predictions are phase-locked to the same cycles as seizures. If false positives cluster in the same cycle phases as true seizures, this further supports the hypothesis that these periods represent genuine proictal states, even in the absence of overt seizures.

2 Methods

2.1 Dataset Description

Data from two different datasets called "Ultra2" and "Competition" are used. Patients are referred to with the initial letter of their dataset, followed by their ID number, e.g., U17 or C03.

Both datasets consist of seizure annotations and electroencephalography (EEG) data, stored as European Data Format (EDF) files. The seizure's starts, and in some cases the type and end, were contained in the annotations.

2.1.1 Ethics Votum

2.1.2 Device Description

2.1.3 Origin of Seizure Annotations

2.1.4 Inclusion Criteria

For a patient to be included in this study, generally speaking, they needed to have enough EEG data and seizures. These were necessary, so the models have enough examples to learn features which can be used to forecast seizures and for the evaluation to carry statistical weight. Only patients who had completed the U002 study at the point of writing and thus whose entire EEG data was available were considered.

Patients needed to have at least 10 valid seizures. In this context, a seizure is considered valid if no other seizure occurs within its preictal interval (section 2.4.3), i.e., a seizure must occur at least ~65 min after the previous seizure.

Regarding the recording duration, there are two separate concepts. First, there is the total monitoring timespan, which is the duration from the start of the first recording to the end of the last. However, data is not recorded all the time, since the device is regularly taken off for a while. So secondly, the total duration of all available recordings is taking into account. Based on these, we want to quantify how much of the monitoring timespan data is available. We define this so-called **recording coverage** as:

$$\text{recording coverage} = \frac{\text{total recorded duration}}{\text{total monitoring timespan}}$$

Patients with a recording coverage less than 60 % were excluded. No patients needed to be excluded on the basis of the total recorded duration, as no patients with little EEG data had enough seizures.

Table 2.1 shows which patients were included and why.

Patient		Seizures			Timespan			
ID	Incl.	Total	Valid	Met	Monitoring	Recorded	Coverage	Met
C01	✓	92	58	✓	494	388	78 %	✓
C02	✓	54	53	✓	537	487	91 %	✓
C03	✓	63	63	✓	415	394	95 %	✓
U01	✓	259	195	✓	401	381	95 %	✓
U02	×	2	2	×	293	102	35 %	×
U03	✓	59	52	✓	394	295	75 %	✓
U04	✓	64	62	✓	197	180	91 %	✓
U05	✓	76	68	✓	358	299	84 %	✓
U06	×	0	0	×	93	84	90 %	✓
U07	✓	13	13	✓	132	124	94 %	✓
U08	×	9	7	×	179	137	76 %	✓
U09	×	0	0	×	301	291	97 %	✓
U10	×	0	0	×	201	165	82 %	✓
U11	×	0	0	×	216	155	72 %	✓
U12	✓	37	30	✓	309	199	64 %	✓
U13	×	4	4	×	202	115	57 %	×
U14	×	5	3	×	420	398	95 %	✓
U15	✓	30	23	✓	352	308	88 %	✓
U16	✓	16	16	✓	346	222	64 %	✓
U17	✓	21	21	✓	264	248	94 %	✓
U18	×	4	4	×	210	188	89 %	✓
U22	×	3	3	×	3	3	89 %	✓

Table 2.1: Patient Inclusion: shows which inclusion criteria were met for which patients. Next to the valid seizures (≥ 10) and recording coverage (≥ 60 %) conditions is denoted whether they were met.

2.2 Dataset Cleaning

The initial datasets had many formatting inconsistencies and erroneous data. This section will outline which issues were found and how they were dealt with.

2.2.1 Seizure Annotations

The annotations of the Ultra2 dataset were initially stored as raw text files, generally with one file per combination of patient and visit.

These files suffered from inconsistent formatting and naming, typos, and files being placed in the wrong patient's directory. To systematically work with the files, these issues were rectified, and they were converted to a comma-separated values (CSV) format for clearer structure and combined into a single file per patient.

Patient U03 had seizure annotations during a period which was no longer available, as the data was lost during the study. These annotations were removed.

2.2.2 EEG Recordings

For some of the EEG files, the following systematic issues were identified:

1. The file was corrupted.
2. The file was labeled as "unknown implant" and contained only artifacts.
3. The file reported a start date in the early 2000s, i.e., before the start of the study.
4. Duplicate copies of a patient's files were found in another patient's directory.
5. Duplicate copies were present within a patient's directory.
6. Multiple files shared the same start timestamp within a patient's directory.

Files with issues 1–3 were removed. For files with issues 4 and 5, only a single copy in the correct directory was retained. For files with issue 6, the longest file was retained; if multiple files had the same length, one was selected at random.

Additionally, it was verified that all EDF files have the same sampling frequency, the same channel names in the same order, and all annotated seizures were within recordings.

2.3 Overview of the Processing Pipeline

After dataset cleaning, the analysis pipeline begins with a comprehensive preprocessing stage. This includes timestamp localization to ensure all EEG recordings are temporally aligned, accounting for time zone differences and daylight saving time (DST) changes. Artifacts in the EEG signals, such as extreme amplitude events or technical noise, are systematically detected and removed to prevent contamination of downstream analyses. The continuous EEG data is then segmented into short, fixed-length windows (*segments*) and longer, non-overlapping windows (*clips*), with careful labeling of intervals relative to seizure events (e.g., preictal, interictal). The dataset is partitioned into training and test sets, ensuring sufficient representation of both seizure and non-seizure periods for robust model development and evaluation.

Feature extraction is performed on each segment, yielding a set of quantitative descriptors that capture both temporal and spectral properties of the EEG. These include signal variance, autocorrelation function width (ACFW), Pearson correlation between channels, and absolute band powers across standard frequency bands. Some features are used as direct inputs to machine learning models, while all are retained for subsequent cycle analysis. The extraction process is designed to maximize the interpretability and physiological relevance of the features, supporting both predictive modeling and the investigation of cyclical patterns in seizure risk.

Two main model architectures are employed: a CNN that operates on raw EEG segments, and an ensemble of feature-based multilayer perceptrons (FB-MLPs) that utilize the extracted features. The CNN architecture is adapted to the specific characteristics of the dataset, including its sampling rate and channel configuration, and is regularized to prevent overfitting. The FB-MLP ensemble leverages z-score normalized features and aggregates predictions across multiple independently initialized models. Both models are trained using only preictal and interictal segments, with careful handling of class imbalance and standardized training procedures to ensure comparability.

Model evaluation is conducted using both standard machine learning metrics (such as area under the curve of the receiver operating characteristic (ROC AUC), precision, and sensitivity) and event-based metrics (EBMs) tailored to the clinical context of seizure prediction. The latter include measures like time in false warning (TIFW) and event-based (EB) sensitivity, which reflect the real-world utility of the models in predicting seizures while minimizing unnecessary alarms. Thresholds for issuing warnings are optimized to balance sensitivity and specificity, and all evaluations are performed separately on training and test sets to assess generalization.

To further understand the temporal dynamics of seizure risk, cycle extraction is performed on the feature time series. This involves gap filling, outlier handling, and bandpass filtering to isolate multidien (multi-day) oscillatory components. The relationship between cyclical features and event timing is quantified using circular statistics, including phase locking value (PLV) and the Rayleigh test for phase preference. A non-parametric permutation test is used to compare the phase distributions of seizures and model-predicted false positives (FPs), with multiple testing correction applied to control the false discovery rate (FDR).

Finally, the propensity evaluation framework integrates these analyses to determine whether a model's FPs are meaningfully aligned with periods of increased seizure risk. Qualification criteria are defined for each patient, feature, and model, requiring significant phase locking and predictive performance. Only when these criteria are met are the phase relationships between FPs and seizures interpreted as evidence of the model capturing cyclical seizure propensity, supporting the overall objective of understanding and predicting seizure risk in a clinically relevant manner.

2.4 Preprocessing

2.4.1 Timestamp Localization

EDF files store their start as a naive timestamp, i.e., without information about their time zone. DST caused the timestamps to have different UTC offsets within a patient. To rectify this, the timestamps for a patient were all converted to that patient's standard time — referred to as their "main" time zone — based on their locale.

The challenge when doing this is that the time shift backwards in autumn creates ambiguous times during the hour that occurs twice. These timestamps' UTC offset cannot be inferred without an ordered sequence of timestamps in which the time jumps back by one hour at some point.

To achieve this for the competition patient's files, the files were sorted by the sequence of recording, and then the file's start and ends were put in order in the format *file1 start, file1 end, file2 start, file2 end, etc*. This allowed the detection of where the end of the previous file was later than the start of the current file, which indicated a time shift backwards, i.e., the shift from DST to standard time. This allowed for finding the border of which files were in daylight saving time and their appropriate conversion to standard time.

For Ultra2 patients, the EDF files contained additional information about their time zone, which was used for conversion.

2.4.2 EEG Artifact Removal

There are two types of artifacts that occur frequently in the EEG data. Their amplitude is very extreme compared to the rest of the signal; thus, they would disrupt multiple parts of the further processing, such as model training and cycle extraction, making it important to remove them.

As shown in fig. 2.1, artifacts usually occur in both channels. The signal hits its minimum possible amplitude of $-1374.21\,\mu V$.

In the first type of artifact, it just clips for one or more samples, as shown in the left panel of fig. 2.1. In the second type of artifact, the clipping is followed by rapid oscillation, then a DC drift before stabilization back to the clean signal, as shown in the right panel of fig. 2.1.

To filter out these artifacts, first, all signals are scanned for clipped intervals where there are samples that are within $0.1\,\mu V$ of the most extreme possible values on either channel. After the end of each clipped interval, the variance is calculated in a two-second window overlapping with a 0.5 sec step. As long as consecutive window's variances are above a threshold, they keep getting removed. The high-variance check captures the oscillations as well as the DC drift. Multiple variance thresholds were tested, and a threshold of $20\,000\,\mu V$ was determined to reliably catch these artifacts without removing valid data. Care was taken not to remove too much high-variance signal, as this often occurs during seizures, which are of particular importance to the analysis. The detected invalid intervals were then padded with 0.25 sec on both sides to ensure that the adjacent data, which might also have been contaminated, are removed. Finally, intervals were merged if they were less than a segment's length apart, since that is the shortest window used here.

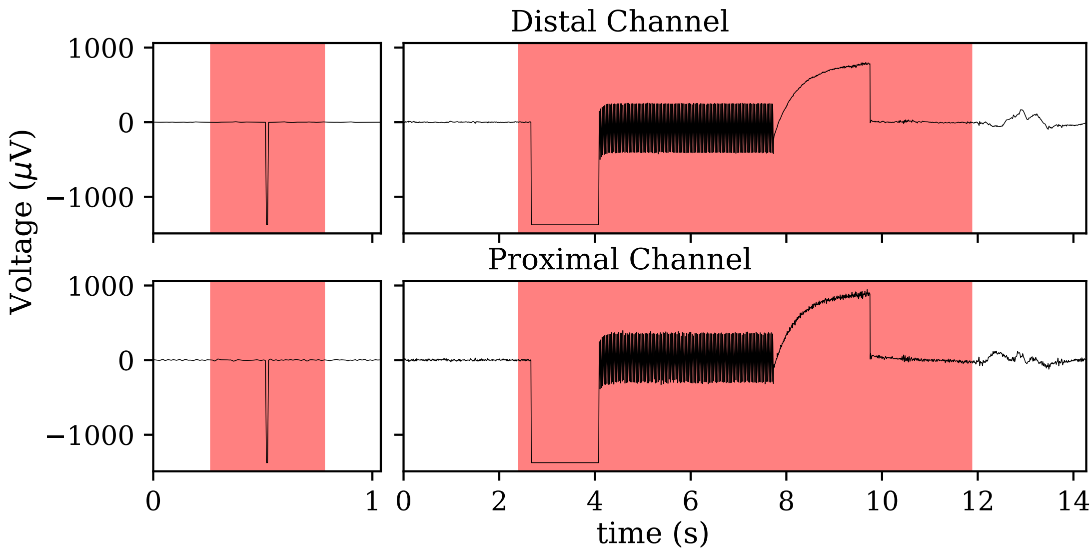

Figure 2.1: Typical EEG Artifacts. Removed intervals are shown in red. Left: A short artifact where the signal is clipped for a single sample. Right: A longer artifact where the clipping is followed by rapid oscillation and a DC drift.

This resulted in a set of invalid recording intervals, which were excluded from the further analysis, as well as a set of valid intervals, which were included.

2.4.3 Windowing

For downstream parts of the pipeline, we create two types of window in the recordings: short *"segments"* and longer *"clips"*. In this paper, these terms carry the specific meanings that are defined in this section.

Segment Duration

Segments are the shortest windows that are used. Based on Yang et al. (2024), they are approximately 15 sec long. However, the EEG data here has an unusual sampling frequency of 207.031 054 658 198 7 Hz, which for a 15 sec segment would result in a fractional number of samples. To avoid this, the exact segment duration was changed to the closest duration that resulted in an integer number of samples which resulted in a duration of 14.997 75 sec with exactly 3105 samples.

The timespan from the start of the first valid interval to the end of the last is completely filled with contiguous, non-overlapping segments, resulting in an aligned series of segments without gaps. Segments that are fully contained in valid intervals are labeled as "existing" and segments that are only partially or not contained are considered non-existing and are not further considered.

Segments receive a "type" which indicates their position relative to close seizures. This type is determined based on the end of a segment, as predictions are issued at the end of clips, which are based on the end of segments.

Interval Types

For seizure prediction to be useful, it is necessary to have a certain minimum intervention interval between the warning time, when the prediction is issued, and the seizure's start. This would enable the administration of an inhibitory drug or the patient to take cautionary action. On the other hand, for a warning to be useful, the seizure must occur within a certain time frame after the intervention time has passed. The window in which the seizure is expected to occur after the intervention time is called the seizure prediction horizon (SPH).

The equivalent to the SPH when looking backwards in time from a seizure, rather than forwards in time from a prediction, is called the preictal interval. It is the window before a seizure where a warning should be issued (fig. 2.2).

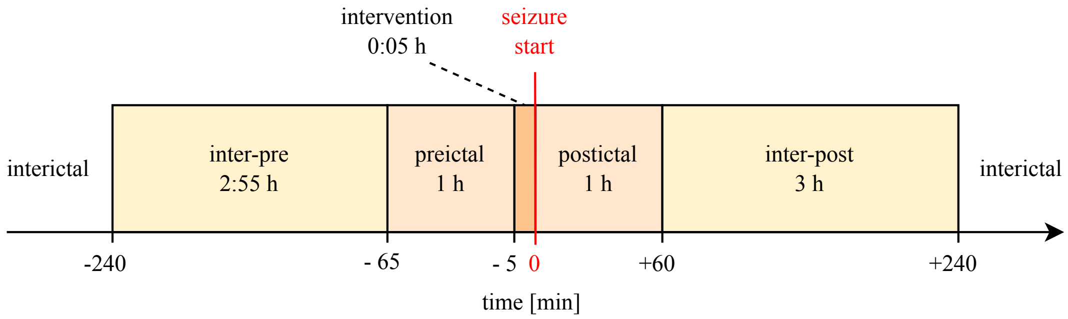

Figure 2.2: The intervals around a seizure with their approximate durations and relative time points. *Not to scale.*

Additionally, it is important to take into account that around the seizure, even beyond the preictal interval, the signal properties might be changed. The period before the preictal interval is labeled as "inter-pre". Similar to the preictal interval, an interval after a seizure is called "postictal" and the period after that is called "inter-post". Periods that do not fall within any of these intervals are labeled as "interictal".

The exact durations of the intervals are calculated as a multiple of the segment duration, again due to the unusual sampling rate (table 2.2). This facilitates analysis by letting each interval have a whole number of segments.

Clips

As in Yang et al. (2024), non-overlapping clips that are approximately 10 min long are created. As with the other intervals, their exact duration is adjusted to a multiple of a segment (table 2.2). The clips are aligned to the segments and usually contain 40 thereof, regardless of whether they are existing segments. They inherit their type from the segments they contain, which means they can have mixed types. Since clips are used for later evaluations, they should not contain a mix of preictal and non-preictal segments, as they could not be clearly labeled then. For this reason, they are also aligned to the ends of preictal intervals, such that one clip always ends at the end of a

Interval	Approximate duration [h]	Segments contained	Exact duration [h]
segment	0:00:15	1	0:00:14.99775
clip	0:10:00	40	0:09:59.91000
inter-pre	2:55:00	700	2:54:58.42500
preictal / SPH	1:00:00	240	0:59:59.46000
intervention	0:05:00	20	0:04:59.95500
postictal	1:00:00	240	0:59:59.46000
inter-post	3:00:00	720	2:59:58.38000

Table 2.2: The approximate and exact durations of the defined intervals, as multiples of segments.

preictal interval, and further clips are created looking backwards in time from there.[1] This means each preictal interval will be fully filled with six clips. Hence, the algorithm that creates the clips takes the end of each preictal interval, and the end of the last valid recording interval as seeds, and creates clips looking backwards in time from there. However, the duration between seizures is usually not divisible by 10 minutes without remainder, and therefore the duration between the ends of preictal intervals is neither. This means that the number of segments between preictal intervals is not usually divisible by 40, and therefore, when looking backwards, the last clips before the next preictal interval will usually be shorter than 40 segments.

This results in two possible issues for a clip:

1. It contains too few segments, whether they exist or not.
2. It contains too few existing segments, i.e., it is not sufficiently located in a valid recording interval.

Clips that do not meet these criteria are considered invalid and excluded from parts of the later procedure. Regarding the first issue, a clip must contain the full 40 segments to be valid. Regarding the second issue, a clip must contain at least 35 existing segments to be valid. This lets clips be that contain short artifacts or whose edge falls outside a recording, but excludes clips where a significant portion is missing.

2.4.4 Dataset Partitioning

We want to train the models on a subset of the data and evaluate their performance on unseen data. To achieve this, the data is split into a train and test set. Since predictive models must use past data to predict future events, the time series is split at a certain time point, rather than allocating individual clips to the sets. An additional reason for this is that we evaluate cycles of up to 50 days, which requires long stretches of continuous data.

Generally speaking, for partitioning, it is important that the train set contains enough samples of each class to learn the patterns and the test set must contain enough samples to accurately

[1] Since this is also done for the test set, it is one of the reasons why the performance evaluation cannot be considered purely prospective as with unseen data.

evaluate the performance. Here, to evaluate cycles, the test set must also be long enough to include at least the longest cycle.

The classes here are preictal and non-preictal. Since seizures only create ~1 h of preictal data but only tend to occur every few days or less, there is a large class imbalance in the data with scarcity of preictal and an abundance of non-preictal data – only 0.76 % of all patient's clips are preictal. Hence, it is important to consider the number of seizures per split, in addition to the recording duration. We aimed to allocate ~60 % of seizures for training and ~40 % for testing.

The train-test boundary was determined per patient as a timestamp on the clip grid. First, seizures were ordered chronologically and an initial split was targeted at 60 % training seizures and 40 % test seizures. If N denotes the number of valid seizures, the number initially allocated to training was $\lfloor 0.6N \rfloor$. The approximate split timestamp was set to the midpoint between the last training seizure and the first test seizure. This timestamp was then aligned to the clip representation by snapping it to a clip start.

Next, the split was constrained using valid clips to ensure adequate EEG data in both partitions: the train partition had to contain between 40 % and 70 % of valid clips. If the snapped split violated these bounds, the split position was shifted in clip-index space to the nearest admissible bound. Finally, the boundary was required to start at an interictal clip; if necessary, the closest interictal clip was selected (searching forward or backward depending on which bound adjustment was needed). The start time of this clip was used as the final boundary.

Patient U03 had a large gap in the middle of his monitoring timespan where data was lost during the study. Following the partitioning procedure, the boundary would have fallen slightly before this gap, creating a very short portion of data, followed by a large gap. This hindered the cycle evaluation; therefore, the boundary was manually set to fall into the gap for this patient.

2.4.5 Feature Extraction

The features extracted from the signals per segment as shown in table 2.3. Some of these were used as inputs to the FB-MLPs, and all were used for the evaluation of cycles.

Feature	#Values per channel	#Values	Used for ensemble
Correlation coefficient	-	1	✓
ACFW	1	2	
Variance	1	2	
Bandpowers	5	10	✓
Total		15	

Table 2.3: Features and their properties

We computed the signal variance and ACFW based on Maturana et al. (2020), and Pearson's correlation coefficient between the two channels, and the spectral band powers of each channel's signal based on Müller et al. (2022).

The ACFW was calculated for each channel. The autocorrelation function (ACF) represents the correlation of a signal s with itself shifted by a lag value λ, which corresponds to a number of samples. $\bar{s}$ is the signal mean, T is the number of total samples, Var_s is the variance of s.

$$\mathrm{ACF}_\lambda = \frac{1}{T} \frac{\sum_{t=1}^{T-\lambda} (s_t - \bar{s})(s_{t+\lambda} - \bar{s})}{\mathrm{Var}_s} \tag{2.1}$$

The ACF is bounded between -1 and 1. The ACFW is defined as the lag λ where the ACF reaches its half-maximum value of 0.5. It can be interpreted as a measure of temporal persistence: it quantifies how quickly a signal decorrelates over time.

Lastly, the spectral band powers were calculated for the delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–35 Hz), and gamma (35–100 Hz) bands for each channel. Band powers were computed for each segment and channel from the power spectral density (PSD) estimated with Welch's method.

Specifically, the signal was divided into Hann-windowed subwindows of length $(2/f_{\mathrm{min}})$ (where $f_{\mathrm{min}} = 0.5$ Hz is the lower bound of the lowest spectral band), with 50 % overlap, detrending by subtraction of the mean, and density scaling (resulting in the unit $\mathrm{V}^2/r\mathrm{Hz}$). This yielded a PSD over frequency for every segment-channel pair.

For each predefined band, the absolute band power was then obtained by numerically integrating the PSD between the band limits using Simpson's rule, with the frequency-bin spacing as the integration step. It was important to use the absolute, rather than relative band power, so that it stays comparable between segments. The result is one scalar band-power value per band and per channel (10 values total for two channels and five bands).

2.5 Model Architectures

In this project, a CNN and an ensemble of FB-MLPs were trained per patient to predict seizures. This section will discuss their exact specifications.

2.5.1 Convolutional Neural Network

In this project, the architecture of the CNN was adapted from Müller et al. (2022) which used the nv1x16 topology originally proposed in Eberlein et al. (2018). It analyzes a segment and uses the raw EEG signals per EEG channel as inputs. Until the final stages, it treats EEG channels separately, computing features per channel. Only at the end are the per-channel features combined to output a final score.

In the following, EEG channels refer to electrode-derived input signals, whereas CNN channels refer to convolutional feature maps (tensor depth).

Figure 2.3 shows the exact architecture. The inputs are passed through a combination of convolutional and pooling layers. There are two distinct types of convolutional blocks: conv1 and conv2. Conv1 uses a convolutional layer, followed by a pooling layer with a relatively large kernel size, reducing the height (#samples) rapidly. All convolution layers use L1 and L2 regularization with a penalty of 10^{-9}. Four conv1 blocks are followed by three conv2 blocks. These further reduce the height by using valid padding for two convolutional layers and pooling again. Ultimately, the sample dimension (height) is reduced to 1 with 32 features per EEG channel. Only then are the per-channel features combined (flattened). These pass through two final dense layers to create the output logit, which is then bounded between 0 and 1 with the sigmoid function, to create the final output.

Since the height gets reduced further and further in deeper layers, the extracted features receive information from more samples, i.e., the receptive field of deeper neurons grows on the time scale. Since the per-channel features are only combined after the final conv block, information from the EEG channels is evaluated separately until the end.

To accommodate two differences in the data structure, compared to Eberlein et al. (2018), two adjustments were made to the architecture. Here, 2 EEG channels are used instead of 16. This means there are 8 times fewer nodes in the first flattened tensor. For this reason, the number of nodes in the first dense layer was reduced by a factor of 8: from 64 to 8. The flat tensors thus have a height of $64 \rightarrow 8 \rightarrow 1$, instead of $512 \rightarrow 64 \rightarrow 1$.

Secondly, because the sampling rate of ~207 Hz here differs slightly from the sampling rate of 200 Hz, the number of input samples from a segment differs too (3105 vs. 3000). Thus, to arrive at a height of 1 at the final conv layer, it is necessary to reduce this dimension slightly more. For this reason, the pool padding of the second conv2 layer was changed from same to valid. This minor change is enough to fully accommodate the sampling frequency difference.

2.5.2 Ensemble of Feature-Based Multilayer Perceptrons

The second model is an ensemble of FB-MLPs that was also adapted from Müller et al. (2022). Instead of raw signal, it uses hand-picked features: specifically the band power spectrum and correlation coefficient from section 2.4.5. This allows it taking features of individual channels and their correlation into consideration.

An individual FB-MLP uses the z-score normalized features as its input layer. It has three dense, hidden layers comprising 16, 8, and 4 nodes, which use recfitied linear unit (ReLU) as an activation function. Batch normalization follows the input and hidden layers. The output layer uses the sigmoid activation function.

The ensemble consists of 50 FB-MLPs that are initialized with different random weights and share their input layer. The outputs of the individual models are then averaged to get the final output. Müller et al. (2022) use an ensemble size of 100, rather than 50. However, they also use many more features as inputs, which enables greater variability between models. Since the variability cannot be as high here, a smaller ensemble size was chosen to reduce resources needed for training.

2.5.3 Model Training

To train the models, only preictal and interictal segments were used. Inter-pre and intervention segments were not used as are shortly before a seizure and thus might often contain similar features to preictal segments. If this was the case, they would make the model learn the wrong patterns since they are still labeled as 0 (not preictal). Postictal and inter-post segments were not used because they might be in the preictal window of non-valid seizures and thus contain preictal features. Additionally, post-seizure, the brain state can be altered significantly, causing changes in the signals, which could confuse the models (Müller et al. 2022).

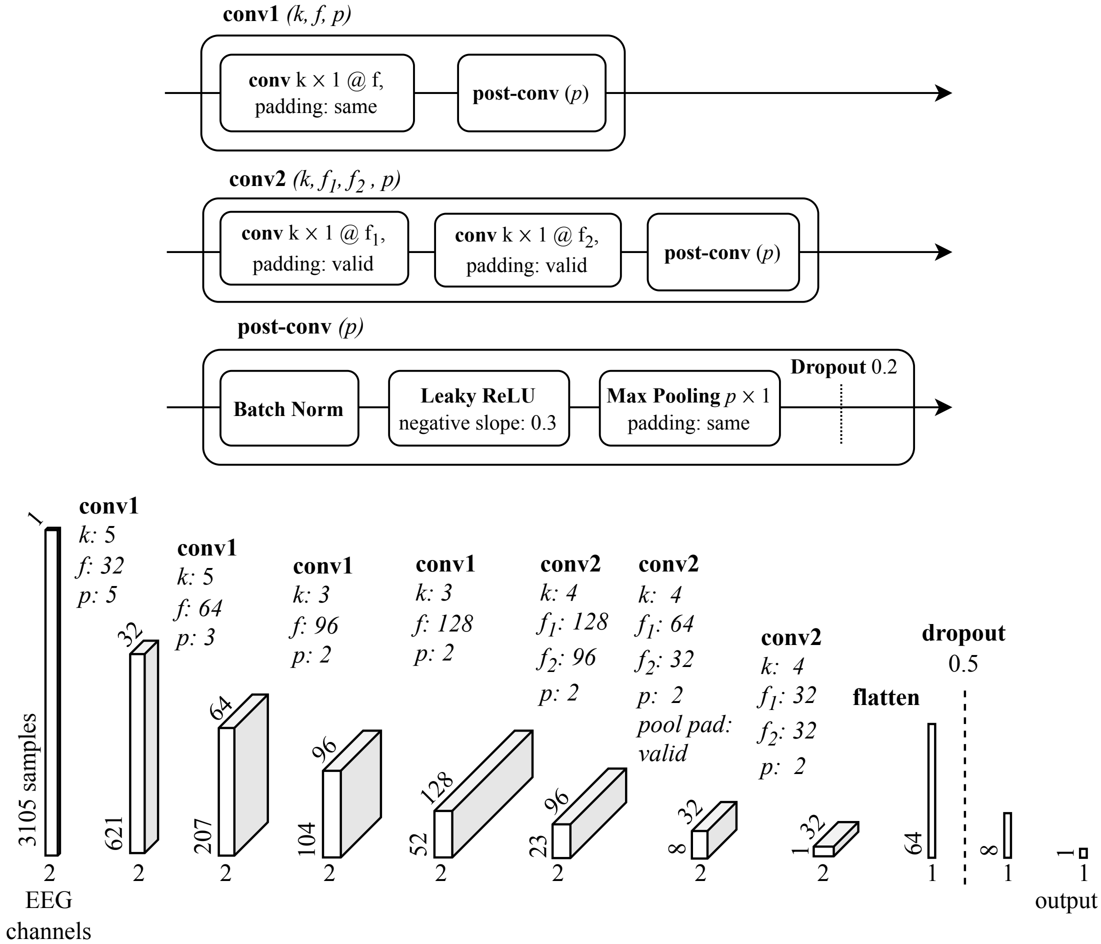

Figure 2.3: CNN Architecture.
Top: Types of special blocks where *k*: kernel size *f*: #filters, *p*: pool size.
Bottom: A segment's EEG data is processed from left to right. The volumes represent the input tensor and the outputs of layers. In between the volumes, the network's layer's properties are shown.

As mentioned in section 2.4.4, there is a large class imbalance with scarcity of preictal segments. To not waste resources training on more interictal segments than necessary, we subsampled the number of interictal segments to max. 20 times the number of preictal ones. The class weights were adjusted to the final ratio of preictal vs. interictal segments, so that the models were not biased towards focusing mostly on learning to predict interictal segments. All models received the same subsampled segments for training, so that results are more comparable.

As is Eberlein et al. (2018), the CNNs were trained for 50 epochs with ADAM optimization, a learning rate of 0.001, and binary cross-entropy as a loss function.

To train the FB-MLPs, before subsampling, a z-score normalizer was fit to the features of the entire train set. This same scaling was then always applied when this model was later used, also for the test set. They were then trained for 200 epochs using stochastic gradient descent for optimization, a learning rate of 0.0001, and binary cross-entropy as a loss function. Müller et al. (2022) used 500 epochs for training. However, here the performance could not be seen to improve after 200 epochs, which is likely also due to the smaller input dimension, requiring less fitting.

2.6 Model Evaluation

After training, it was important to evaluate the performance of the models on the valid clips, as conclusions about the propensity of seizures could only be drawn from the FPs, if the models were decent at predicting seizures. FPs from a random or worse-than-chance classifier would not carry the intended meaning.

For the ensemble, the input features were scaled again using the z-score normalizer, which was previously fit on the train data. Then, the models were used to issue scores (0–1) for all existing segments, which are the model's prediction of how likely that segment is to be preictal. The segment scores were then averaged per clip to get the clip scores. These, rather than segment scores, are used for the further evaluations.

We used two types of metrics to evaluate the performance: clip-based metrics and EBMs.

Here, clip-based metrics refer to the standard measures used to evaluate models in machine learning on samples (clips), which use the confusion matrix of true positives (TPs), FPs, false negatives (FNs), and true negatives (TNs) to compute metrics, such as precision, sensitivity, accuracy, and the ROC AUC. The Hanley McNeil test (Hanley and McNeil 1983) was performed to test for statistically significant improvement of the ROC AUC compared to a random classifier.

On the other hand, EBMs are specific to time-series data and seizure prediction and quantify more clinically relevant properties. Here, the (relative) TIFW (Mormann et al. 2007; Eberlein 2025) and the (relative) number of seizures predicted were used.

2.6.1 Time in False Warning

Relative TIFW is the EB equivalent to the false positive rate (FPR), i.e., the inverse of specificity. That is to say it is minimized when non-preictal (false) samples are correctly classified as such. However, when a prediction is issued from a ~10 min clip, it is not issued for the timespan of that clip, but for that clip's SPH, i.e., the timespan ~5–65 min after the clip's end. This means that summing the duration of FP clips is not sufficient to calculate the TIFW. The clip's SPHs and overlaps thereof must be taken into account. When a system unnecessarily alarms the patient, it causes them distress and inconvenience, which is why it is essential to minimize false alarms.

To calculate TIFW, only valid, non-preictal clips are selected. For all such clips that issue a warning (FPs), the SPHs are selected. Overlapping SPHs are then merged, so that shared time is not counted multiple times. The total duration of these non-overlapping false-warning intervals is the absolute TIFW. The relative TIFW is obtained by normalizing absolute TIFW by the maximum possible warning time, which is the sum of the non-overlapping SPH intervals from all non-preictal clips. This yields a metric bounded between 0 and 1 which enables comparable interpretation across patients and is consistent with the guidelines in Mormann et al. (2007).

2.6.2 Event-Based Sensitivity

It is also essential to quantify the relative number of seizures successfully predicted, which is also referred to as EB sensitivity. For preictal clips, a positive prediction is considered a true warning, and a seizure is counted as predicted if it falls within the SPH of at least one positively predicted preictal clip. This yields the number of predicted seizures, and EB sensitivity is then computed as the ratio of predicted to total seizures. This metric therefore reflects clinically relevant event detection performance at seizure level rather than clip level.

2.6.3 Threshold Optimization

Whether a clip issues a warning depends on whether its score is greater than or equal to a classification threshold. To optimize it, metrics are calculated for all possible thresholds, i.e., all unique model scores. To take both the EB specificity (1 - rel. TIFW) and the EB sensitivity into account, the harmonic mean between them is calculated:

$$H(a, b) = \frac{2ab}{a + b} \tag{2.2}$$

$$S_{\mathrm{EB}} = H\big(\mathrm{Sensitivity}_{\mathrm{EB}},\ 1 - \text{rel. TIFW}\big) \tag{2.3}$$

The threshold that maximizes this EB score is set as the best threshold and used to determine binary clip predictions in further steps.

The metrics are calculated for the train and test set independently. The results for the test set yield a more accurate evaluation of the performance, since it is unseen data. However, the optimal threshold is determined for the test set individually, rather than using the train set's threshold. This again cannot be considered purely prospective performance, but improves the performance, which is an advantage for the cycle extraction, and the priority here.

2.7 Model Comparison

If different models both classify a portion of data as preictal, even though no seizure occurs afterward, this is interpreted as evidence for a proictal state, since preictal features were apparently present in the data. To quantify the tendency of false predictions to co-occur across models, we calculated the correlation of prediction errors (COPE) (Müller et al. 2022) between the ensemble and CNN scores for each patient. Its definition follows.

Let $\vec{p}_i$ and $\vec{p}_j$ be the model's outputs for models i and j, respectively, and $\vec{L}$ the correct label vector. The prediction error vector of model i is defined as $\vec{e}_i = |\vec{p}_i - \vec{L}|$ (analogously for model j). The weight w_k for each sample k, the weighted mean $\langle z \rangle_w$, and weighted standard deviation $\sigma_w(z)$ are defined as:

$$\begin{aligned} w_k &= \max(e_{i,k}, e_{j,k}), \\ \langle z \rangle_w &= \frac{\sum_k w_k z_k}{\sum_k w_k}, \\ \sigma_w(z) &= \sqrt{\langle z^2 \rangle_w - \langle z \rangle_w^2}. \end{aligned} \tag{2.4}$$

The COPE is then

$$c_w = \frac{\left\langle \left(\vec{e}_i - \langle \vec{e}_i \rangle_w\right) \left(\vec{e}_j - \langle \vec{e}_j \rangle_w\right) \right\rangle_w}{\sigma_w(\vec{e}_i)\, \sigma_w(\vec{e}_j)}. \tag{2.5}$$

This measure emphasizes coherence in false predictions by assigning higher weight to samples where at least one model has a large prediction error. Consequently, samples with low errors contribute little, whereas samples with high errors dominate the estimate.

2.8 Cycle Extraction

We aim to identify and quantify the rhythmic patterns of seizure occurrences and predictions in the time series data. This section describes the methods used to preprocess the data, handle missing values, extract relevant events, and perform circular statistical analyses to relate cycles in features to the timing of seizures and model predictions.

2.8.1 Gap Filling

Cycle extraction requires gap-free data, so gaps are filled before processing. This is done on the segment level because more data are available there than at the clip level. Otherwise, segment data from invalid clips would be lost.

The gap-filling procedure preserves the original segment timeline while imputing only short missing intervals (< 14 d) in the extracted feature series. Long gaps (≥ 14 d) are treated as structural recording interruptions and are therefore not imputed. Instead, they are used to split the data into independent continuous chunks, ensuring that donor values are never drawn across major recording breaks.

Within each fillable gap, imputation is performed independently per feature column by local donor sampling. For a gap of length L, donor candidates are taken from neighboring segments before and after the gap within a window of size $W = \max(L, N_{\min})$, where $N_{\min} = 100$ is the minimum donor span. Values are sampled with replacement from valid donor rows and assigned to the missing positions. Additionally, each sampled value is perturbed by a multiplicative factor in the interval $[1 - \delta,\ 1 + \delta]$, where $\delta = 0.5$ is the maximum distortion percentage. This design preserves local feature statistics, avoids unrealistic interpolation over long outages, and introduces slight stochastic variation to reduce deterministic copy artifacts in the imputed sequences.

2.8.2 Events

Cycle extraction is based on two types of events: seizure onsets and FPs predicted by the models. Only FPs from valid clips in the test set are included, determined using the optimal threshold from section 2.6.3. Seizures from the entire dataset are used, as some patients have fewer than 10 in the test set. This is justified because seizure timing is independent of model training, under the assumption that phase locking between seizures and features is stable across the dataset.

2.8.3 Outlier Handling

Feature-based phase locking can be affected by artifacts in the EEG signal, such as muscle or technical noise. To mitigate this, extreme feature values are clipped at the 99.999 % quantile. Various thresholds were tested for all patients and features, and this one was visually determined to remove the extreme outliers while preserving the vast majority of the data. After clipping, features are z-score normalized to standardize their distributions.

2.8.4 Filtering

A non-causal, multidien filter is applied to each continuous feature chunk to isolate slow oscillatory components. Specifically, a 10th order Butterworth bandpass filter in second-order-section (SOS) form is used with a period range of 5–50 days, and a sampling frequency corresponding to the segment duration. This process extracts multidien dynamics while removing short-term fluctuations.

2.8.5 Phase Locking

Based on the filtered features, the phases at event times, PLVs, and mean angles are computed using the Hilbert transform. The PLV quantifies the consistency of event timing relative to the phase of an oscillatory process, ranging from 0 (no phase preference) to 1 (perfect phase locking). This is a standard tool in circular statistics and neuroscience for quantifying phase-event relationships.

Computation of PLV

1. **Analytic Signal:** Given a real-valued time series $x(t)$ (e.g., a filtered feature), the analytic signal $z(t)$ is computed using the Hilbert transform:

$$z(t) = x(t) + i \cdot \mathcal{H}[x(t)] \tag{2.6}$$

   where $\mathcal{H}[x(t)]$ is the Hilbert transform of $x(t)$.

2. **Event Phase Extraction:** The instantaneous phase is then:

$$\phi(t) = \arg(z(t)). \tag{2.7}$$

   For each event (e.g., a seizure / FP), extract the instantaneous phase at the event times where N is the number of events: $\{\phi_{e_1}, \phi_{e_2}, \ldots, \phi_{e_N}\}$.

3. **Mean Resultant Vector:** The mean resultant vector is

$$R = \frac{1}{N} \sum_{k=1}^{N} e^{i\phi_{e_k}}. \tag{2.8}$$

4. **PLV Calculation:** The PLV is the magnitude of the mean resultant vector:

$$\mathrm{PLV} = |R| = \left| \frac{1}{N} \sum_{k=1}^{N} e^{i\phi_{e_k}} \right| \tag{2.9}$$

5. **Mean Angle:** The argument of R gives the preferred phase:

$$\mu = \arg(R) \tag{2.10}$$

2.8.6 Rayleigh Test

The statistical significance of phase locking is assessed using the Rayleigh test for each feature, which tests the null hypothesis that the event phases are uniformly distributed.

2.8.7 Circular Comparison

To determine whether two groups of circular data, i.e., event phases from two event types, have significantly different mean directions, a non-parametric permutation test is used. This is appropriate for circular data, where standard linear tests are not valid due to the periodic nature of the data. The null hypothesis that both samples are drawn from populations with the same mean direction is tested.

Let $\{\phi_1^{(1)}, \ldots, \phi_{n_1}^{(1)}\}$ and $\{\phi_1^{(2)}, \ldots, \phi_{n_2}^{(2)}\}$ be the two samples of event phases (in radians), with sample sizes n_1 and n_2.

Observed Statistic: First, compute the circular mean for each group:

$$\bar{\phi}^{(j)} = \arg\left(\sum_{k=1}^{n_j} e^{i\phi_k^{(j)}} \right), \quad j = 1, 2 \tag{2.11}$$

The observed difference in mean direction is then

$$\Delta_{\mathrm{obs}} = \mathrm{wrap}(\bar{\phi}^{(1)} - \bar{\phi}^{(2)}) \tag{2.12}$$

where $\mathrm{wrap}(\cdot)$ denotes wrapping the angle to the interval $[-\pi, \pi)$.

Variance and Test Statistic: To account for the concentration of the data (i.e., how tightly phases are clustered), the test uses a variance estimate based on the mean resultant length R_j for each group:

$$R_j = \frac{1}{n_j} \left| \sum_{k=1}^{n_j} e^{i\phi_k^{(j)}} \right| \tag{2.13}$$

The variance of the mean direction for each group is approximated as

$$\mathrm{Var}_j = \frac{1 - R_j}{n_j(R_j^2 + \epsilon)} \tag{2.14}$$

where ϵ is a small constant to avoid division by zero. The test statistic is then

$$T_{\mathrm{obs}} = \frac{|\Delta_{\mathrm{obs}}|}{\sqrt{\mathrm{Var}_1 + \mathrm{Var}_2}} \tag{2.15}$$

This statistic is analogous to a Welch's t-test, but adapted for circular data.

Permutation Procedure: To generate the null distribution, the two samples are pooled and randomly reassigned into two groups of sizes n_1 and n_2, without regard to their original labels. For each permutation, the difference in circular means and the corresponding test statistic T_{perm} are computed as above. This process is repeated a large number of times ($N_{perm} = 5000$), yielding a distribution of T_{perm} under the null hypothesis.

P-value Calculation: The two-sided p-value is calculated as the proportion of permutations where the permuted test statistic exceeds or equals the observed test statistic:

$$p = \frac{1 + \sum_{i=1}^{N_{\mathrm{perm}}} \mathbb{I}(T_{\mathrm{perm},i} \geq T_{\mathrm{obs}})}{1 + N_{\mathrm{perm}}} \tag{2.16}$$

where $\mathbb{I}(\cdot)$ is the indicator function and N_{perm} is the number of permutations.

Bootstrap Confidence Intervals: In addition to the permutation test, bootstrap resampling is used to estimate confidence intervals for the mean directions and their difference. This involves repeatedly resampling each group with replacement, recalculating the circular means, and determining the empirical percentiles of the resulting distributions.

Interpretation: A small p-value indicates that the observed difference in mean direction is unlikely to have arisen by chance, suggesting a significant difference in the phase preference between the two groups. The bootstrap confidence intervals provide an estimate of the uncertainty in the mean directions and their difference.

Usage: The permutation test is applied to all pairs of events: seizures vs. CNN FPs, seizures vs. ensemble FPs, and CNN FPs vs. ensemble FPs.

The Watson-Wheeler test, which tests for equality of circular distributions, was also considered as an alternative but found to be overly sensitive to event phase variability.

2.8.8 False Discovery Control

Since the Rayleigh and circular tests are performed for each feature (15 p-values per test), results are corrected for FDR using the Benjamini-Hochberg (BH) procedure, yielding FDR-adjusted p-values for each test.

2.9 Model-Feature Propensity Evaluation

To rigorously evaluate whether a model's FP predictions reflect underlying phase-dependent seizure propensity, we define a set of qualification criteria for each *patient, feature,* and *model*. These criteria ensure that only meaningful model-feature relationships are interpreted in the context of cyclical seizure risk.

2.9.1 Qualification Criteria

For a given patient, feature, and model, the following conditions must be satisfied for significance level $\alpha = 0.05$:

1. **Seizure-Feature Phase Locking:** The patient's seizures must exhibit significant phase locking to the feature, as determined by the Rayleigh test with BH FDR correction ($p_{\text{Rayleigh,BH}} < \alpha$). This criterion ensures that the feature is even relevant to the patient's seizure timing.
    - Again, due to limited seizure counts, seizures from both training and test sets are included, under the assumption that phase locking remains stable across the dataset.
2. **Model Predictive Performance:** The model must achieve a test set ROC AUC ≥ 0.75. This empirically chosen threshold ensures that only models with acceptable predictive accuracy are considered.
3. **FP-Feature Phase Locking:** The model's FPs must also be significantly phase locked to the feature ($p_{\text{Rayleigh,BH}} < \alpha$). This indicates that FPs are not randomly distributed in phase.
4. **Distributional Similarity:** If the above criteria are met, the phase distributions of FPs and seizures are compared using the non-parametric permutation test (see section 2.8.7). A non-significant result ($p_{\text{perm,BH}} \geq \alpha$) indicates that FPs and seizures may arise from the same phase-dependent process.

2.9.2 Interpretation of Outcomes

Ideal Case: Strong Evidence for Phase-Dependent Propensity

- Both seizures and FPs: $p_{\text{Rayleigh,BH}} < \alpha$
- Permutation test: $p_{\text{perm,BH}} \geq \alpha$

Interpretation: FPs occur at similar preferred phases as seizures, suggesting the model captures phase-dependent seizure propensity.

Contrary Cases

- For seizures, $p_{\text{Rayleigh,BH}} \geq \alpha$: seizures are not significantly phase-locked to the feature. Hence, any phase locking of FPs cannot be interpreted as a meaningful proxy for propensity.
- For model FPs, $p_{\text{Rayleigh,BH}} \geq \alpha$: FPs are not significantly phase-locked to the feature. The model-feature combination cannot be used to evaluate propensity.
- Permutation test: $p_{\text{perm,BH}} < \alpha$: The phase preference of seizures and FPs differs significantly. FPs cannot be interpreted as occurring during a proictal period.

As such, it is evaluated for each patient, model, and feature combination whether it indicates that FPs are indicating high propensity.

3 Results

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